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(b) Explain how bias can be dected and measured during AI development and deployment.

Bias can be detected during both AI development and deployment.

1. Compare sekdion Rates: check wether different demographic groups are selected or rejeeted at significantly.

2. Compare Accuracy: Measure whether the AI make carrect predictions equally.

3. Check False positive and false Negative Rate.

4. Test the Training Data: check wether the training data represents different groups.

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5. Use fairness Metrics : Metrics such as demographic parity.

6. Regular Monitoring After Deployment:

Bias should not be checked only before lanuch. They should be continuouly monitored because real-world data and user behavior can change over time.

: 7min

⏲: 7min IT23039

(12) Accountability when an Autonomous System causes Harms.

(a) Discuss the Concept of Accountability in autonomous systems.

In this given case, the Autonomous delivery vehicle made an incorrect decision and damaged property.

1. Manufacture: The manufacture is responsible for ensuring that the vechele and AI System are properly designed.

2. Software Team: The Software team is responsible for developing.

3. Manggment: Management may be responsible for decisions about the model.

4. System operator / owner: The operator may be responsible for using and maintaining the system.

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(b) Identify the responsibilities that may be associated with developers testers:

1. Developers:

- Design the AI system properly.
- Write reliable and safe software.
- Indentify and report possible risks.
- Fix known software problem.

2. Testers:

- Test the system carefully under different conditions.
- Identify bugs, failure, and saftey risks.
- Report serious problems before deployment.
- verify that fixes work correctly.

🕒 : 7min

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13: Ethical leadership in IT organizations:

(a) Explain the concepts of ethical leadership in IT organization.

Main features of Ethical ~~Real~~ Leadership –

1. Honesty : Leader should Communicate truthfully.

2. Responsibility : Leader should take responsibility for the effects.

3. Respect for employess: Employee should be able to respont security.

4. User Safe and privacy: Leader should given importance to security, privacy.

5. Fair-Decision-Making: Leader should make decision fairly.

: 7min

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(b) Discuss how leaders can establish an ethical culture through policies.

IT Leader can create an ethical culture by making ethical behavior an important part of everyday work.

1. Policies : create clear policies for security

2. Incentives : Reward employees not only for releasing product quickly.

3. Training : provide regular training on Software ethics.

4. Communication : Encourage employee to open discuss ethical concerns.

5. Transparency : Leader Should be honest about project Risks.

: 7 min

(14)(a) Explain the concept of informed consent in personal data collection.

For consent to be properly informed, user should know:

1. What data is being collected - Such as name, location, contacts or browing information.

2. Why the data is being collected - The specific purpose of collecting it.

3. How the data will be used - Inculding wether it will be analyzed or shared.

4. Who will receive the data.

5. How long the data will be stored.

6. Their choies and rights.

Ⓛ : 3 min

14 (b) Discuss wheather simply obtaining a user's agrements to a lengthy privacy plolicy:

For consent to be meaningful, users should be able to understand what they are agreeing to and make a genuine choice.

Reasons:

1. Too much Information.

2. Unclear Language.

3. Important Information may be Hidden.

4. Lack of Real Choice.

5. Lack of Understanding.

: 5 min

16. (a) Compare copyright and patents as forms of intellectual-property protection.

Copyright vs patents:

Copyright	Patents
(i) original creative expression	(i) New and useful invention
(ii) Source code, GUI design, documentation	(ii) New technical image compression technique.
(iii) Prevent unauthorized copying of the work.	(iii) Give exclusive right over a qualifying invention.
(iv) Original work must be created	(iv) Invention must meet legal requirements;
(v) Thousands of lines of source code and GUI	(v) New image-compression techinque.

: 7 min

(b) Explain which assets are primarily associated with copyright protection.

In this case, different assets can be protected in different way.

Copyright protection:

1. Source Code: The thousands of lines of original Source code are protected.

2. Graphical user Interface (GUI): originals creative elements of the GUI.

3. Documentation: Original Software manuals, written documentation.

Patent protection:

The new image-compression technique may be suitable for patent protection.

🕒: 7min

17(a) Define the digital divide and distinguish between differences in access, skills, quality of access.

The Digital divide has several forms:

1. Difference in Access: Some people have computers, and internet connection while others may have no device or no internet access.

2. Difference in skills: Even if people have internet access, they may not have the digital skills.

3. Difference in Quality of Access: People may have internet access, but the speed, reliability, cost or availability may be different.

: 7min

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(b) Discuss the social implications of the digital divide for education, employment, healthcare

1. Education: Student without reliable internet devices or digital skills may have difficult attending online class.

2. Employment: Many jobs require online applications, digital skill and internet access.

3. Healthcare: Online healthcare information, appoinments.

4. Government services: If government services are available only online.

5. Economic opportunity: Digital technology provides opportunities for online business ess.

: 7min

18. (a) Define green computing and explain it importance.

Green computing means using computers and other information technologies is an environmentally responsible and energy-efficient way.

Importance of Green Computing:

1. Reduces Energy Consumption: Energy-efficient computers and data center.

2. Reduces Electronic Watse: proper recycling, reuse and responsible disposal.

3. Reduces operating costs: Using less electricity can reduce electricity bills.

4. protects the Enviroment: Lower energ use can reduce green-house gas emissions.

5. Encourage sustainable Technology: Its promotes longer device lifetimes, repair.

⏲ : 7min

(C) Explain how Software design decisions can indirectly affect energy consumption.

Software design can indirectly affect energy consumption because software control how much work a computer, server or mobile device performs.

1. Efficient Algorithm: Efficient Algorithm ca requires less processing.

2. CPU usage: Software that continuously uses the CPU can consume more power.

3. Memory Usage: Software that uses excessive memory may require more system resource.

4. Network usage: Applications that frequently transfer learge amount of data.

5. Data Centers: Efficent software can reduse the workload on servers.

: 7min

19.(a) Define conflict of interest in the IT work-place.

A conflict of interest occurs when a persons personal's interest, relationships or benefits can influence - or appear to influence - their professional decisions.

Example: If an employee is choosing a soft software supplier and one of the supplier is owned by their family member, the employee should disclose the relationship to management.

Why it is a problem -

- The decision may not be completely impartial.
- other vendors may be treated unfairly.
- The organization may lose trust in the sel process.

🕓 : 8min

(b) Explain why an actual conflict, potential confli and perceived conflict can all be ethically Significant.

An actual conflict, potential, or perceived conf of interest can all be ethicaly important.

1. Actual conflict : An actual conflict exists when a person's personal interest directly affects their professional decision.

2. potential conflict : A potential conflict

exists when a personal relationship or interest could influence a fune decision.

3. Perceived conflict: A perceived conflict exists when other people reasonably belive that personal interest could influence.

: 7 min

20. (a) Identify the ethical, legal, professional privacy, and cybersecurity issues in the Scenario.

The scenario contains several important issues because the company wants to lanunch the Smart-home platform. even

1. Ethical Issue: Management is prioritizing business pressure and investor expections

over the Safety and interest of users.

2. Legal Issue: The company may have legal Obligations to protect user's personal data.

3. Professional Issues: Software engineers have a professional responsibility.

4. Privacy Issues: Smart-home actitivity data can reveal senitive information about People's daily routines.

5. Cybersecurity Issues: The system has a Vulnerability that allows unauthorized users to access data.

⏲ : 7 min

(C) Identify relevant principles from professional code of ethics.

Several principles from professional code of Such as the ACM/IEEE Engineering code of ethics.

1. Public Interest: professionals should act in public interest.

2. Avoid Harm: Engineers should avoid action that may cause harm.

3. Privacy: Professionals should respect and protect users personal and household data.

4. Honesty and Trustworthines: Engineers Should honestly communicate the security

problem to management and should not hide or misrepresent the risk.

5. Security and Quality : Software should developed and released with appropriate security.

🕒 : 7min